FROM ACCESS TO USE: STRENGTHENING RESPONSIBLE ANTIBIOTIC USE IN NIGERIA.

Addressing self-medication, inappropriate antibiotic use and barriers to professional healthcare.

NGHI Policy Brief No. 01
Published: August 2026
Publisher: NextGen Health Initiative
Author: Olawuwo Idris Adeniyi

1. INTRODUCTION

The inappropriate use of antibiotics remains a major driver of antimicrobial resistance (AMR) in Nigeria, threatening the effectiveness of life-saving medicine and increasing the burden of infectious diseases. This concern is reflected in ongoing efforts by the Nigeria Centre for Disease Control and Prevention (NCDC) to evaluate and strengthen Nigeria's AMR surveillance system. On July 15 2026, the NCDC annouce that its 10-year review of Nigeria's AMR surveillance system has been completed. While the review highlighted progress in Surveillance, it also identified persistence gap that continue to undermine Nigeria's response to AMR.

AMR refers to a situation where microorganisms become resistance to antimicrobial medicines that were previously effective against them. One of the major contributors to AMR is the inappropriate use of antobiotic. AMR is also a one Health issues because resistant microorganisms and antibiotic residue can move between humans, animals, and environment. Therefore, addressing AMR requires appropriate measure from these areas.

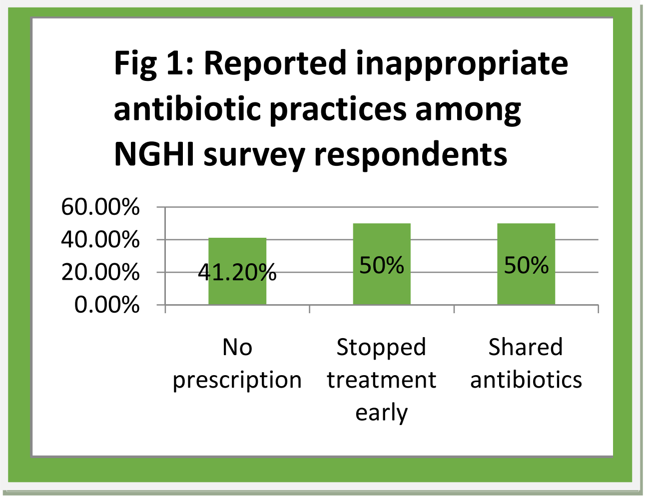
NIGERIA'S CURRENT SITUATION

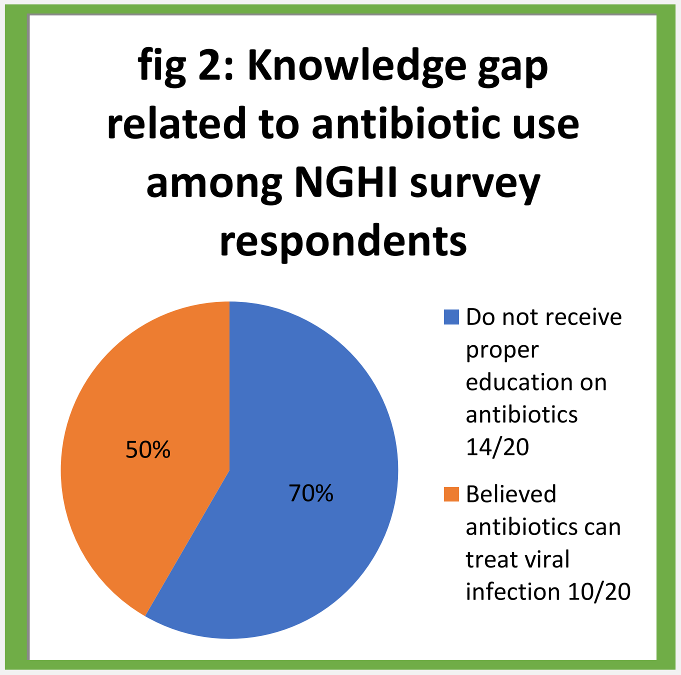
Nigeria has a high burden of antimicrobial resistance (AMR). The country ranks 19 out of 204 countries with the highest age-standardised mortality rate per 100,000 population associated with AMR. Nigeria recorded an estimated 50,500 deaths directly attributable to AMR and 227,000 deaths associated with AMR in 2021, with a significant burden among children under five. The number of deaths associated with AMR was higher than deaths from several major disease groups, including respiratory infections, tuberculosis, maternal and neonatal disorder, neglected tropical diseases and malaria. An implementation review of National Action Plan (NAP 1.0) showed that only 44% of planned activities were completed, highlighting challenges in achieving AMR control targets. In response to the gap identified, the government launched the second National Action Plan on AMR (2.0) in 2024, covering 2024-2028. The new plan focuses on strengthening AMR surveillance, improving awareness and behaviour, infection prevention and control, responsible use of antimicrobials, and cordination among human, animal and environmental health sectors. However, limited laboratory coverage affect Nigeria's ability to detect emerging resistance patterns, generate reliable surveillance data and guide appropriate antibiotic use. Nigeria has only 14 sentinel laboratories in the human health sector, 6 in animal health and 2 in environmental health. This limited laboratory coverage makes it difficult to detect, monitor and respond to AMR effectively across the one health sectors.

Several studies conducted in Nigeria have reported high levels of inappropriate antibiotic practices including self medication and non prescription use. A systematic review and meta-analysis involving 91 studies estimated the pooled prevalence of self-medication in Nigeria at 67.8%, with antibiotics among the most commonly self medicated medicines. Factors contributing to this practices include perceived minor illness, financial barriers, previous treatment experience and easy assess to medicine.

Beyond these existing findings, an ongoing survey conducted by NextGen Health Initiative (NGHI) assessing responsible antibiotic use among Nigerians identified important gaps in antibiotic practices and awareness. Among the 20 respondents who provided consent, 17 (85%) reported using antibiotic in the last 12 months. Of these, 7 (41.2%) reported using antibiotics without prescription, while 10 respondent (50%) reported stopping antibiotics treatment before completion and sharing antibiotics with others, 14 respondent (70%) reported not having proper education on antibiotic use.

These priliminary findings suggest that challenges surrounding AMR are not related to antibiotic availability but also involve how antibiotics are accessed and used by individuals. As these findings are still ongoing, and the numbers of respondents is currently small, they should not be taken as representative of Nigerian population. However, this early findings provide insight into possible areas requiring further investigation and targeted health education.





POLICY PROBLEM

Nigeria has made efforts to curb antimicrobial resistance (AMR), but several problems continue to hinder these efforts, especially the inappropriate use of antibiotics. It is not that there is unavailability of antibiotics, but some people use antibiotics without a prescription, stop treatment before completion, or share antibiotics with others. There is also a gap in knowledge and awareness. In the ongoing NextGen Health Initiative (NGHI) survey, about 70% of respondents reported that they had not received proper education on antibiotic use, while some believed that antibiotics can treat viral infections. Easy access to antibiotics without proper professional guidance can also contribute to misuse. Some people obtain medicines based on previous experience or recommendations from friends and family rather than appropriate professional advice. Other challenges include limited AMR surveillance. Gaps in laboratory coverage and surveillance can limit the ability to detect resistance patterns, monitor emerging threats and generate timely evidence for effective intervention.

These challenges highlight the need for stronger implementation of Nigeria's National Action Plan on AMR (NAP 2.0), which covers 2024–2028.

2. WHY THIS MATTERS

Antimicrobial resistance is a serious public health problem that should not be ignored. If left unchecked, infections that are normally treatable could become harder or even impossible to treat, increasing the risk of severe illness and avoidable death. This could contribute to higher mortality, especially when effective treatment options become limited. As a result of antimicrobial resistance, infections can become harder and more expensive to treat. A person who would normally recover with a commonly used antibiotic may instead require alternative treatment, additional medical care or longer hospitalisation. These can place an additional financial burden on individuals, families and the healthcare system.

AMR can therefore have a wider economic impact. More money may be spent on treatment, hospital care and other healthcare needs, while longer hospital stays can also affect patients, families and healthcare resources.

3. CURRENT POLICY

Nigeria has taken steps to address antimicrobial resistance (AMR) through the establishment of various policies and regulatory measures aimed at preventing self-medication, promoting responsible antibiotic use and strengthening AMR control.

One of the major efforts is the One Health National Action Plan on Antimicrobial Resistance (NAP 2.0), which was launched in 2024 and covers the period 2024–2028. The plan focuses on six strategic objectives, including governance, education and awareness, AMR surveillance, infection prevention and control, antimicrobial stewardship, and research and development.

In addition to the National Action Plan, Nigeria also has regulatory measures aimed at controlling access to medicines, including prescription requirements for certain medicines such as antibiotics.

However, evidence from studies conducted in Nigeria on antimicrobial resistance and self-medication shows that inappropriate antibiotic use remains a challenge. Some individuals continue to obtain and use antibiotics without proper prescription, while gaps in knowledge and awareness about AMR still exist.

This suggests that while Nigeria has established policies to address AMR, challenges remain in implementation, public awareness and ensuring responsible antibiotic use.

4. POLICY OPTIONS

Based on the identified challenges, the following policy options could strengthen Nigeria's response to antimicrobial resistance (AMR) and promote responsible antibiotic use.

A. Expand AMR surveillance through increased sentinel laboratory capacity

Nigeria currently has limited sentinel laboratory coverage across the One Health sectors, with 14 laboratories in human health, 6 in animal health and 2 in environmental health. Expanding these laboratories will improve the ability to detect antimicrobial resistance patterns, monitor emerging threats and provide evidence for effective decision-making

C. Improve access to healthcare professionals and medical advice

Addressing inappropriate antibiotic use requires more than increasing healthcare facilities. Some individuals avoid seeking healthcare because of challenges such as long waiting times, limited availability of healthcare professionals and difficulty accessing medical advice.

Increasing the availability of healthcare workers and improving healthcare delivery systems can reduce these barriers and encourage more people to seek appropriate care before using antibiotics.

B. Shift from general awareness campaigns to targeted antibiotic education

Public education on AMR should go beyond simply telling people not to misuse antibiotics. It should focus on helping people understand when antibiotics are needed, why professional medical advice is important before antibiotic use and the risks associated with self-medication.

Education should also address misconceptions, such as the belief that antibiotics can treat viral infections. Social media platforms should be actively used to deliver simple, accessible and evidence-based information because they provide an opportunity to reach a larger audience.

D. Strenghten enforcement of prescription only antibiotics

Although Nigeria has regulations requiring prescription control for certain medicines, including antibiotics, but enforcement remains a challenge. Strengthening enforcement among pharmacies, patent and proprietary medicine vendors and other medicine outlets can help reduce access to antibiotics without appropriate medical guidance.

5. RECOMMENDED POLICY ACTION

A. Expand AMR surveillance through increased sentinel laboratory capacity

NCDC, in collaboration with the Federal Ministry of Health and Social Welfare (FMoHSW), and State Ministries of Health, should expand the existing sentinel laboratory network by identifying states and areas that currently have little or no access to AMR testing and prioritising them for new or upgraded laboratories. Existing laboratories should also be provided with the necessary equipment, reagents, trained laboratory personnel and funding needed to carry out AMR testing consistently.

NCDC and relevant State Ministries of Health should also ensure that these laboratories are properly connected to the national AMR surveillance system so that laboratory results can be reported and used for decision-making. This will make it easier to identify where resistance is increasing, understand the antibiotics that are becoming less effective, and allow health authorities to respond more quickly to emerging AMR threats.

B. Shift from general AMR awareness to targeted antibiotic education

The NCDC, in collaboration with FMoHSW, State Ministries of Health and relevant community health organisations, should ensure that the National Action Plan on AMR (NAP 2.0) education and awareness campaigns go beyond simply telling people what AMR is. It should focus more on the real effects and dangers of inappropriate antibiotic use, especially self-medication. So, the campaigns should focus on practical situations people face, such as using antibiotics for colds and malaria, buying antibiotics without a prescription, stopping treatment when they feel better, sharing antibiotics with others, or using leftover antibiotics. This would make the education more useful than simply telling people that "AMR is a problem."

When people understand that self-medication and inappropriate antibiotic use can put their health at risk, they may be more willing to seek proper medical advice instead of simply taking antibiotics because they are afraid of becoming more sick or because they want quick treatment.

C. Improve access to healthcare professionals and medical advice

FMoHSW, in collaboration with State Ministries of Health and relevant health authorities should employ more healthcare professionals, especially in facilities where there are many patients, to reduce the long waiting time before patients can see a doctor or other healthcare professionals. In addition, the government should work more on strengthening Universal Health Coverage (UHC) so that people can have better access to affordable and quality healthcare. Findings shows that some respondents reported behaviours that may be linked to financial or access barriers. If someone is sick and knows that they cannot afford to see a healthcare professional, or that they may have to wait for many hours before getting medical attention, they may decide to self-medicate or buy antibiotics directly from a medicine seller instead.

Relevant professional bodies should also ensure regular antibiotic stewardship training for healthcare professionals to improve their knowledge of appropriate antibiotic use and help them provide patients with timely and accurate information about when antibiotics are needed and how they should

D. Strengthen enforcement of prescription-only antibiotics

The Pharmacists Council of Nigeria (PCN), National Agency for Food and Drug Administration and Control (NAFDAC), and relevant State regulatory authorities should strengthen the enforcement of the existing rules on prescription-only antibiotics. Pharmacies, patent and proprietary medicine vendors, and other authorised medicine outlets should not sell antibiotics to anyone without a valid prescription from a qualified healthcare professional. They should therefore increase monitoring of medicine outlets and ensure that outlets found selling antibiotics without a prescription face appropriate sanctions.

If people know that they cannot easily buy antibiotics from a medicine store without a valid prescription, they may be more likely to visit a clinic or hospital to see a healthcare professional before using antibiotics. This can reduce self-medication and give patients the opportunity to receive the right medical advice before taking antibiotics.

6. CONCLUSION

Antimicrobial resistance remains a serious public health concern in Nigeria, and inappropriate use of antibiotics is one of the issues that needs more attention. Nigeria already has policies and strategies in place to address AMR, including NAP 2.0, but gaps remain in surveillance, access to antibiotics without prescription, public understanding of responsible antibiotic use, and access to healthcare professionals.

The preliminary findings from the NGHI survey also show that inappropriate antibiotic use and knowledge gaps exist among some respondents. Although these findings cannot be used to represent the Nigerian population, they highlight behaviours that require further attention.

Nigeria therefore needs to strengthen the implementation of existing measures while focusing more on responsible antibiotic use. Expanding AMR surveillance, enforcing prescription requirements, moving towards targeted antibiotic education, and improving access to healthcare professionals can help reduce inappropriate antibiotic use and protect the effectiveness of antibiotics for future generations.

REFERENCES

1. Federal Ministries of Agriculture And Food Security, Environment, And Health And Social Welfare; April 2024. One Health Antimicrobiall Resistance National Action Plan 2024-2028
2. Institute for Health Metrics and Evaluation (IHME). (2023). The burden of antimicrobial resistance (AMR) in Nigeria. Global Research on Antimicrobial Resistance (GRAM).
3. NextGen Health Initiatives preliminary survey 2026. Knowledge, Attitudes, and Practices on Antibiotic Use Among Nigerians. https://ee.kobotoolbox.org/x/I7ppvWr7
4. Nigeria Centre for Disease Control and Prevention. (2026, july 15). Nigeria just completed a 10-year review of its antimicrobial resistance surveillance system [facebook post]. https://www.facebook.com/100064649291568/posts/1468469441984685/?app=fbl
5. Muoka CG, Akparanta NV, Taribo FA, Onubogu CO, Asiegbu DA, Azubike CJ, Nwokoro IO, Athan CE, Uwaechina El, Ede U. Self-medication practices in Nigeria: A systematic review and meta-analysis. Niger Med J 2026; 67 (4): 1271-1296. https://doi.org/10.71480/nmj.v67i4.1358
6. World Health Organization 2024 (NAP 1.0 and 2.0)